



Kattirsitti Jeet Govindrao
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Communication & Signal Processing

22B0010
Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 31/07/2004

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2027	
Intermediate	SSC	Vishwaganga Jr.Sci. College	2022	81.33%
Matriculation	CBSE	Podar international school, Latur	2020	95.40%

Pursuing a **Dual minor** in **Centre for Machine Intelligence and Data Science (CMINDs)** and **Computer science (CS)**

SCHOLASTIC ACHIEVEMENTS

- Secured a **branch change** from Aerospace(Btech.) to **Electrical (Dual degree)** [2023]
- Secured a rank of **2111** in **JEE Advanced** out of **0.14 M+** aspirants [2022]
- Attained a percentile of **99.80** in **JEE Mains** among **1M+** candidates [2022]
- Scored full (**100/100**) in **Information technology** in **CBSE Class X Examinations** [2020]
- Qualified for level-2 of **Dr. Homi Bhabha Balvaidnyanik Competition** (conducted by Greater Bombay Science Teachers' Association). [2019]
- Secured **Gold medal** in **NSO** (National science olympiad) in **X** conducted by SOF. [2019]

INTERNSHIPS AND EXPERIENCES

GMRT | Research project

[ongoing]

Guide : Kaushal Buch

- Conducted a review of prior work on **eGMRT**, focusing on **higher resolution** and **adaptive nulling**.
- Developed an interactive **MATLAB** tool to detect faulty **Focal plane array** FPA elements.
- Simulated antenna performance for **Digital beamforming** via the **phased array toolbox** in **MATLAB**.

KEY PROJECTS

Computer Vision | Self project

[May-Jun2024]

- Implemented image manipulation algorithms like blurring, sharpening, **Sobel**, **edge detection** and thresholding using **OpenCV** library, having useful applications in **contactless biometrics**.
- Implemented **Face detection** algorithm, converted images to grayscale to prepare for face detection.
- Employed the **Haar Cascade Classifier** to detect faces in images, further used in facial recognition.

Recommender systems | Self project

[Jun-Jul2024]

- Boltzmann machine** - Developed a binary recommender system for movie ratings, employing a **probabilistic generative model** that learns underlying patterns in user preferences and item characteristics.
- Implemented a custom class based on leveraging **Gibbs sampling** and **contrastive divergence**.

RNN LSTM architecture| Self project

[Jun-Jul2024]

- Sentiment analysis** - Implemented an **RNN** for sentiment analysis on movie reviews and classify them as positive or negative. Successfully tested in **Postman** to simulate **API requests** and responses.
- Spam detection** - Implemented a robust **NLP** model for spam email filtering. successfully tested the machine learning model using .pkl files in **Postman** to simulate **API requests** and responses.
- Google stock prices** - Implemented an architecture which leverages **LSTMs'** ability to capture long-term dependencies in sequential data, making it well-suited for **time series** forecasting tasks.
- Despite the randomness of stock prices, successfully captured both **upward** and **downward** trends. Tested its accuracy by simulating **API requests** and responses on two months of data using **Postman**.

QNA bot deployed via gradio using Google FLAN| Self project

[Jun-Jul2024]

- Developed a chatbot leveraging a **780M** parameter Google FLAN for enhanced conversational capabilities.

- Enhanced a **gradio GUI** to interact with the **LLM** model, improving user interaction and usability.
- Implemented a **pipelined GPT-2** text-to-text transformer AI model with **top-k** sampling and **temperature controls**, allowing users to adjust hyper-parameters to influence the generated output.

Supervised and unsupervised learning | Self project

[May-Jul'24]

- Programmed an **ANN** for classifying an image as pizza or not_pizza with **adversarial training**. Developed a **CNN** for classifying images using **TensorFlow** libraries and dropout layer regularization.
- Implemented a **Self-organizing map** for **fraud detection** based on maximum **mean inter-neuron distance** as a metric for outliers and frauds on a bank's credit card approval dataset.

Miscellaneous projects

[Apr'23-Jul'24]

- Created a **Line follower bot** which was designed to negotiate a slope of 10°, 20°, and 30°.
- Instantiated and managed **virtual machines (VMs)** on the **Google cloud engine** for data processing.
- Leveraged **Apache Spark's API Pyspark** for distributed data processing on **NSE dataset**. Utilized **DBeaver** for SQL database management and querying.
- **PID** controller gain tuning using a Neural networks in **MATLAB** for non-linear system.

COURSE PROJECTS

Multicycle Processor Design

[November'23]

Guide Prof: Prof. Virendra singh, Department of Electrical Engineering, IIT Bombay

- Designed a multicycle datapath using VHDL for the instruction set of a 16-bit computer named IITB-CPU.
- Modeled an **FSM controller** using behavioral modeling and performed RTL simulations on datapath.

IITB-RISC 16-bit processor

[Apr'24-May'24]

Guide Prof: Prof. Nikhil Karamchandani, Department of Electrical Engineering, IIT Bombay

- Design a **6-stage pipelined processor**, IITB-RISC-23, based on **Little computer architecture**.
- Implemented **Hazard mitigation units** for reducing the penalty cycles for dependent instructions.

ECG

[April'24]

Guide Prof: Sandeep Mondal, Department of Electrical Engineering, IIT Bombay

- Successfully built an **ECG amplifier** that accurately captured and amplified the heart's electrical signals.
- Implemented a **notch filter** for filtering out 50Hz frequency noise effectively from the ECG signal.

KEY COURSES UNDERTAKEN

- **Electrical Engineering:** Analog circuits, Digital systems, Signal processing -I, Communication systems, Electromagnetic waves, Microprocessor, Electronic devices, Control systems.
- **Data Science DS203** (Programming for Data Science). DS303 (Introduction to machine learning).
- **Other:** Computer Programming and Utilization, Biology, Economics, Introduction to Aerospace Engineering, Doing Sociology, Introduction to cryptography and number theory, A historical perspective of Aerospace Engineering, Cybersecurity**, App development**.

(** Online courses Udemy)

EXTRA-CURRICULAR ACTIVITIES

- Secured Top rank in **NAC** conducted by STEM RESEARCH & INNOVATION (New Delhi) [2017]
- Secured a **certificate of merit** in **Green olympiad** conducted by **TERI**. [2019]
- Participated in **EnB Buzz** organized by EnB Club and worked in a team of three to come up with a sustainable **business model canvas** for a **Secret heirloom** company [Dec'22]
- Participated in **Vigilance awareness week competition** conducted by the Vigilance Cell of the Institute on the Topic of **Corruption Free India: Steps Ahead for Developed India** [Oct'22-Nov'22]
- Studied **Excel and big data handling** and **Latex** under **Learners' Space**. [2023]
- Completed a year long training under NSO (**National sports organization**). [2022]
- Completed **Natural language processing NLP-Deep learning** bootcamp. [2024]